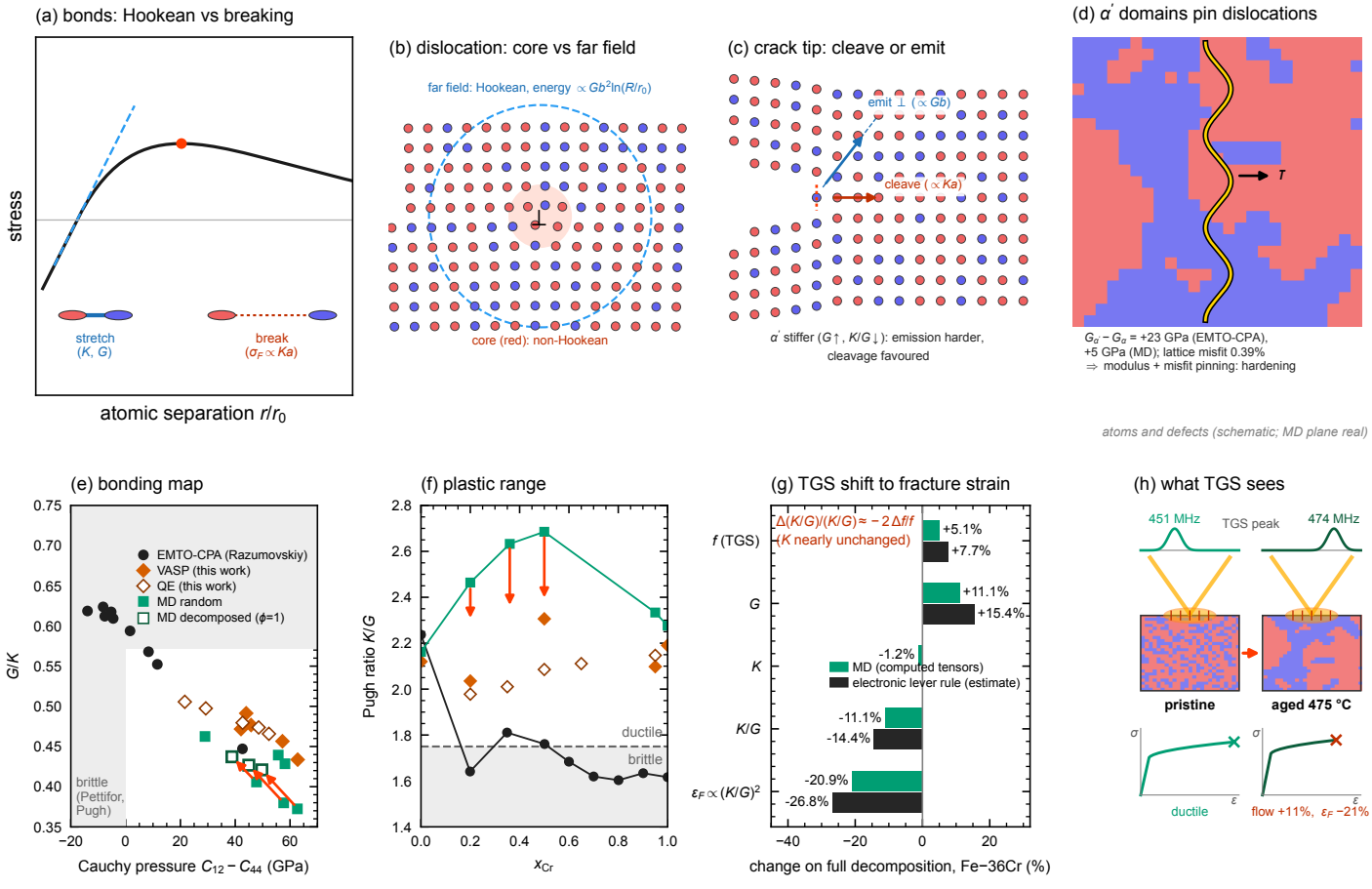

(a) bonds: Hookean vs breaking
stress
stretch
(K, G)
break
(σF ∝ Ka)
atomic separation r/r0
(b) dislocation: core vs far field
far field: Hookean, energy ∝ Gb2ln(R/r0)
core (red): non-Hookean
(c) crack tip: cleave or emit
emit ⊥ (∝ Gb)
cleave (∝ Ka)
α′ stiffer (G↑, K/G↓): emission harder, cleavage favoured
(d) α′ domains pin dislocations
T
Gα − Gα = +23 GPa (EMTO-CPA), +5 GPa (MD); lattice misfit 0.39%
⇒ modulus + misfit pinning: hardening
atoms and defects (schematic; MD plane real)
(e) bonding map
G/K
EMTO-CPA (Razumovskiy)
VASP (this work)
QE (this work)
MD random
MD decomposed (ϕ=1)
brittle (Pettifor, Pugh)
Cauchy pressure C12 − C44 (GPa)
(f) plastic range
Pugh ratio K/G
ductile
brittle
xCr
(g) TGS shift to fracture strain
Δ(K/G)/(K/G) ≈ −2Δf/f (K nearly unchanged)
f (TGS)
G
K
K/G
εF ∝ (K/G)2
+5.1%
+7.7%
+11.1%
+15.4%
-1.2%
MD (computed tensors)
electronic lever rule (estimate)
-11.1%
-14.4%
-20.9%
-26.8%
change on full decomposition, Fe−36Cr (%)
(h) what TGS sees
451 MHz
TGS peak
474 MHz
pristine
aged 475 °C
σ
ε
ductile
flow +11%, εF −21%